Cosmological Relativity: what the construction delivers, what it declines, and where its edge is

D. Janzen

Abstract

This paper states, in one place, what the Cosmological Relativity corpus establishes, at the weight each result is established at, together with what the construction declines to claim and where its open edge lies. Nothing here is new: every claim is drawn from one of seventeen papers or from the ledgers those papers cite, and is stated at the register its source gives it—a proposition as a proposition, a reading as a reading, a conjecture as a conjecture. What is new is the assembly, and the two things an assembly can supply that a set of papers cannot: the joins, where a result in one paper is load-bearing for a result in another, and the relative size of the frontier against the body of work that surrounds it.

1 What forces the object, what it is, and the distinctions it forces

Cosmological Relativity is a construction on one geometric object, and almost everything below is that object read from somewhere. But the object is not where the argument starts, and presenting it first would ask a reader to grant the thing in question. It is reached: a measurement forces a cosmic foliation, reading that foliation ontologically is the augmentation, and maximal symmetry then leaves exactly one manifold to carry it. This section takes those in order, and closes on the distinction every later result depends on — what a *boundary* of such an object is.

1.1 What the measurement forces

The cosmological redshift is not a property an emitting source carries. It is the path integral of the expansion rate along the photon's own trajectory, accumulated continuously rather than fixed at emission [15]. That distinction is what makes the isotropy of the microwave background's monopole a measurement of something other than what it is usually taken to measure.

Grant the common reading its strongest form. Let last scattering be *perfectly* homogeneous. Then the observed temperature contrast is exactly the variation in the integrated expansion, with nothing left over, and *the entire observed isotropy is the statement that the integrated expansion was the same in every direction.* Homogeneity at the source does not *supply* the isotropy; granted perfectly, it removes one term and leaves the isotropy as a direct measurement of uniform expansion [15].

Under the alternative — that the large-scale expansion rate tracks the matter lumpiness region by region, with no single global scale factor — the accumulated redshift would scatter across the sky by $\sim 10^{-3}$, against an observed $\lesssim 3 \times 10^{-6}$: *an exclusion by three orders of magnitude. And every choice in the estimate biases it downward, so the number is a floor rather than an estimate* [15].

Two escapes remain and they are closed separately. The *statistical* escape is closed by the floor. The *structured* escape — a finely tuned, spherically symmetric inhomogeneity centred on the observer, which would give an isotropic redshift with no global uniform expansion and is untouched by any $\sqrt{N}$ argument — is closed by the Copernican principle *together with* the independently measured isotropy of the expansion history itself. *With both excluded, the observed isotropy forces uniform cosmic expansion.*

What a single datum thereby establishes is worth stating exactly, because it is larger than a parameter bound. The rejected region is not a corner of parameter space but *the entire class of histories lacking a single global scale factor on a single global time.* To reject that class is to select a global, time-ordered expansion — the cosmic foliation, made dynamical. *And the foliation is logically prior to everything the standard model otherwise assumes*: "space" is one of its slices, "expansion" an ordered family of them, isotropy and homogeneity properties of a slice — none of them statable until the foliation is fixed.

1.2 The augmentation that follows, and its structural counterpart

Augmenting general relativity by fixing a physical foliation and reading it ontologically — the lapse the objective rate at which the existing world advances, the shift the relativity of synchrony — is *necessary and sufficient* for a coherent formal description of an existing, evolving world [16]. *Necessary*, because a description lacking the lapse part collapses existence into occurrence. *Sufficient*, because the two parts close the only coherent escapes. No field equation is altered.

The standard reading fills general relativity's silence about which foliation is physical by collapsing two distinctions at once: it reads the relativity of simultaneity as the absence of any objective present, and it reifies the four-manifold as a thing that exists [15]. §2 takes those apart; the augmentation is what remains when they are held apart.

A second collapse entered with the first cosmology and runs in the opposite direction from what the measurement establishes. Einstein's static solution fixes the cosmic scale from the content — with $\lambda = \kappa\rho/2$ the radius is $R = 1/\sqrt{\lambda}$, so *the density of the world-matter determines the size of the world* — and Friedmann's equations carry that direction of determination forward, the scale factor's evolution sourced by what the universe contains. *What the isotropy forces is the reverse ordering for the rate*: it is fixed independently of how the content is distributed, since a rate tracking the content is precisely what the measurement excludes. *Whether the same reversal extends from the rate to the scale is not settled by the measurement* [15].

And the structure the datum forces was posited, attacked, defended and abandoned within four years. Einstein assumed it in February 1917 in his own words "against the spirit of relativity", on the empirical ground that stellar proper motions are small compared with c. De Sitter countered a month later, placing the choice among candidate universes outside physical argument altogether and pressing as his sharpest objection that in Einstein's solution "time has a separate position". Eddington defended it in 1920 on geometric grounds, in the declared absence of any experimental knowledge on cosmical scales — *he names the absence of the discriminator and selects anyway.* Einstein then went nearly silent on cosmology and never addressed how the cosmic time his own assumption distinguished stands to the relativity of simultaneity [15]. *The assumption was correct, and the objection was correct as a description of the structure though not as a reason to reject it.*

And the same augmentation is forced a second time, from general relativity alone. The event horizon of a collapsing body is a metric singularity met only at infinite exterior time, so no finite layer carries a completed horizon (§1.6) [2]. *These two forcings are independent and co-equal, and neither rests on the other*: one is measured and one is structural, and a reader may take either without the other.

1.3 Three senses of "real", fixed before the word is used

The corpus uses *real* in three senses and keeps them apart; this paper does the same. They are not degrees of one thing.

Real in the arithmetic sense: a geometry has a real coordinate basis, with no imaginary coordinate doing any work. That is the sense in which the substrate below is a real manifold, and the claim is about coordinates and signature and nothing more.

Real by construction: a feature is built by a construction and really seen from the vantage that builds it. The perspectival mass is of this kind, and so is the compact face on which a continuous $\mathfrak{su}(3)$ could live.

Existent: the ontological sense, whose criterion in this framework is temporality. *The existent is the evolving three-dimensional layer, and the manifold is its record* [16]. By that criterion the compact face is real by construction but, being Riemannian and atemporal, *not a co-equal existent*—it holds no clock [17].

So nothing in this section claims that a five-dimensional object is what exists. The substrate is the geometric core the construction reads and cuts; what exists, on this framework's own criterion, is the layer. The higher-dimensional geometry is what the layer is a cut of, not a thing possessing the layer's kind of being, and the four-dimensional manifold is the record rather than the existent.

1.4 What maximal symmetry then selects

The foliation is forced; which manifold carries it is a separate question, and this is where the construction's one object arrives — as a selection rather than a premise. The maximally symmetric solutions of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ form a one-parameter family with curvature radius $\alpha = \sqrt{3/|\Lambda|}$. They are usually met as hypersurfaces in a flat five-dimensional space, which is a convenient picture and not the manifold: solving that embedding for the fifth coordinate and substituting back returns an intrinsic four-dimensional line element in which the manifold is coordinatised by four *real* coordinates, real regardless of the sign of the curvature.

The imaginary fifth coordinate is a property of one embedding picture, not of the surface. *This is reality in the first of the three senses above, and in that one only.*

Reading the metric's eigenvalues off that real line element gives the result the programme's ontology stands on. Three eigenvalues are positive always, and the fourth's sign is fixed by real data alone.

Proposition 1 (The unique intrinsically Lorentzian maximally symmetric manifold)**.** *De Sitter space is the only real maximally symmetric manifold carrying an intrinsic Lorentzian signature. It has a real coordinate basis, satisfies the maximal-symmetry requirement, and is the only such manifold with the requisite causal structure [19, 1].*

Two ways the signature can turn are worth separating, because they are of different kinds. At fixed positive curvature the crossing occurs where the fourth eigenvalue passes through a *pole*: its magnitude diverges and its sign flips, the numerator being constant throughout. *So the metric does not degenerate at the crossing*—a signature change through zero would send the determinant to zero and leave no metric there, and that is not what happens. The one place the eigenvalue vanishes is the null cone: not a crossing within a member of the family, but the family's own singular leaf [19].

That the substrate is reached through imaginary instruments is therefore a fact about the instruments. The embedding coordinate, the equatorial seam's continuation, and the reassignment at the cosmogenesis are each places the construction uses an imaginary variable and lands on a real manifold; the geometry is intrinsically real at every point they land on [19]. In one case the instrument is not optional and the case is the construction's own central algebraic object: over the field of the mass parameter the horizon cubic is irreducible, and below the Nariai mass its discriminant is positive, so its three roots are real and distinct. That is exactly the hypothesis pair of *casus irreducibilis*, whose conclusion is that no root lies in any real radical extension—*the three real horizon radii admit no expression in real radicals as functions of the mass*, and every radical route to them passes through $\mathbb{C}$ [19]. The imaginary is the only available road, and the roots it lands on are real.

1.5 One scale, and what that means

The substrate carries a single dimensionful scale. Read projectively it is a Cayley–Klein geometry[L] whose *absolute* is the null cone, and its geodesic separation is the Cayley–Klein log-cross-ratio taken against that absolute, with the Cayley–Klein constant equal to α [19]. So α is not a length inserted into a metric but the constant of a projective geometry whose absolute is the light cone—which is the signature proposition above, read projectively rather than metrically.

The constants through which physics is written on it are unit gauges, each a nameable feature of the substrate: c is the null-ruling slope of the equilateral hyperboloid, Λ the waist, and G enters only as the offset-length GM/c^2, the mass being the offset of the cut from the central geodesic [19, 12, 4]. On this accounting the gravitational–cosmological–quantum sector spends no free dimensionless constant, and the reason is structural rather than counted: neither real form supplies a second invariant, and a dimensionless magnitude needs two [19].

What the count leaves is not a constant but an initial condition, and the corpus carries it as measured content rather than as a derived quantity—as flat ΛCDM carries the baryon-to-photon ratio from a baryogenesis it does not model [14].

1.6 What a boundary of the substrate is

Here the corpus draws the distinction that everything downstream depends on, and it is a distinction between two kinds of singular locus rather than between a real one and an artefact.

A *metric* singularity is a locus at which a spatial extent contracts to zero while the curvature stays finite. A curvature singularity is a locus at which an invariant of the curvature diverges. The event horizon of a collapsing body is the first kind: a null hypersurface along whose generators the spatial extent also contracts to zero, occurring only as the null future boundary approached at infinite exterior time [2]. Three consequences of that follow immediately, and each is a distinction a reader needs before meeting any application.

Observer-dependence is not the criterion. It is tempting to sort these loci by whether they are seen by every observer, and the sorting fails at once: the Rindler horizon is perspectival and carries a flux, so "perspectival, therefore no flux" is refuted before it starts. What sorts the four horizons the corpus treats is *completion*—whether the boundary is reached on any finite layer—and completion sorts all four [2].

And the finite-curvature case is the one the metric-singularity result treats. It is a different object from the locus standardly called the curvature singularity at $r = 0$, where the invariants computed on the areal radius diverge; the first is established with the event horizon as its exemplar, and nothing is said there about the second beyond noting that it is a different object [2]. *That the two admit a common treatment, and what becomes of the divergence at* $r = 0$ *under it, is established in the companion* [3, 4]. *No claim of necessity or sufficiency is made here in either direction*: the scope is what is stated, and reading it as a general criterion for when a boundary may be crossed would assert what the source declines to assert.

And formal likeness does not sort the two classes. A chart such as Eddington–Finkelstein parametrises approaches to a singular locus by a coordinate that degenerates there, so a single metric place is drawn spread out as an extended line. Mercator draws the North Pole by the same projective mechanism, and the visual appearance is identical: a place drawn as a line. *But the reason for the appearance is the opposite.* At the pole the chart manufactures an extension that is not there, and a chart centred on the pole collapses the line back to the point it always was. At the horizon the chart spreads out a metric collapse that *is* there: reading the line back yields no ordinary point a better chart would reveal, because no chart removes it—*the collapse is in the metric, not in the projection* [3]. So no argument from how a singular locus is drawn can sort the two, in either direction.

1.7 What follows for the reading, and what does not

Two guards belong here rather than later, because both are misread in the same direction.

Perspectival does not mean unreal. Where the corpus calls a feature perspectival it means built by a construction and really seen from the vantage that builds it—the curvature divergence of a perspectival metric is real as that metric's, and reading it as an artefact in the pejorative sense is the misreading the epistemology is written to defeat [6, 3].

And identifying a feature as perspectival is not a dismissal of the construction that carries it. Sweeping a radial curve to recover a spatial geometry is the legitimate and indeed the natural way to chart a geometry from a given vantage, and the result agrees with the standard geometry on every observable outside the horizon. What is identified is not an illegitimate operation but a misreading of its product: treating features the construction manufactures—the asymmetric labelling, and with it the curvature singularity at $r = 0$—as features of the geometry rather than of the chart. *The construction is faithful as a description from its vantage; the error is ontological* [3].

2 What the forcings pry apart

The forcings of §1 do their work by separating pairs of ideas the standard reading holds fused. *Each separation is a result rather than a clarification*: in every case the fused form is shown to make a claim the evidence refuses or the formalism cannot carry, and the separated form is what the rest of the construction is written on. This section states the pairs, because a reader who keeps any of them fused will read §§3–7 as claiming something other than what they claim.

2.1 Literal and perspectival

The appearances divide into components a projection images faithfully and components that are artefacts of the projection, and *both classes are non-empty* [6]. The Sun's annual path along the ecliptic is a perspectival illusion of the Earth's motion; the Moon's monthly circuit is a literal orbit. Because both occur, *no blanket reading is admissible* — and reading every appearance literally is the first thing a method must forbid.

A perspectival component need not be an illusion, and the corpus's own case is the sharp one. The metric's verdict that two null-separated horizon events share one place and one instant is a genuine fact about the measure — the separation truly collapses — *yet perspectival still*, a fact about the ruler and not about the identity of the events, which stay distinct on the point-set the ruler is laid over [2]. *So the perspectival class is not the class of false appearances but of those that image the projection rather than the world*, and some are entirely real as facts about the projection.

Nor does formal likeness sort the two, which is why the separation cannot be done by inspection — §1.6 gives the case, where two loci drawn identically arise for opposite reasons.

2.2 Saving an appearance and explaining it

An admissible world must *exhibit* the projection under which a perspectival appearance arises and show why it appears as it does — *not discard it, and not merely reproduce it* [6]. A model can reproduce the ecliptic path exactly while saying nothing true about why it appears. *The retrograde loops are not real reversals to be tracked by finer machinery but the parallax of the Earth's own motion*, and the right theory dissolves the puzzle by identifying what the appearance is.

And a dissolution can be entirely ontology-free, which is worth having before §5. The continuation through the curvature singularity undoes the standard verdict not by any claim about which manifold is fundamental but by exhibiting the continuation that carries the curve through

— *the puzzle dissolved by identity, nothing assumed about the world beyond the appearance's own smoothness* [3].

2.3 Existence and occurrence

Events occur; the world exists. To deny the distinction — to hold the four-manifold itself the existent — is to treat occurrences as existents, which on analysis smuggles a fifth, meta-temporal dimension along which the block is supposed to "exist", and is incoherent as a basic description [15]. *The regress is the argument*: whatever endures does so through some time, so a four-manifold said to exist requires a further time it endures through.

This is the distinction §1's third sense of "real" turns on, and the lapse is its formal counterpart: *the rate of advance of the layer that exists, which is what a clock measures.*

2.4 Simultaneity and synchrony

The relativity of synchrony is general — a feature of relative motion, the light postulate fixing only its magnitude — and it is the *shift*, the freedom of a moving observer to tilt the slices off the comoving normal. *Simultaneity is the separate question.*

To read the relativity of synchrony as the denial of an objective simultaneity is a modal fallacy [15, 6]. From the premise that the appearances contain no local discriminator between two candidate worlds, it does not follow that the worlds are identical, nor that the structure distinguishing them does not exist: *the absence of a local test is not the absence of the fact.*

The phrase is exact rather than figurative. Locally flat is the technical condition that every point of an embedding has a neighbourhood in which it is standard, and high-dimensional topology turns on locally flat against wild embeddings precisely because local standardness leaves the global invariant free: a submanifold can be locally indistinguishable from the trivial one at every point and globally knotted. *So the modal fallacy is the inference from local triviality to global triviality*, and that inference has a standing counterexample in every dimension where the question has been asked [6].

The history is a record of the fallacy committed and corrected. No stellar parallax was detectable for two millennia and this was read as telling against the Earth's motion — though before the cosmos was given depth the relevant parallax was not a conceivable measurement at all. No local experiment distinguishes rest from uniform motion, and this was read as the non-existence of any preferred rest. *In each case the missing discriminator was mistaken for a missing fact, and a later non-local measurement supplied both* [6]. *An absent discriminator is not an absent fact*, and whether a discriminator is even available is a question about the picture in hand rather than about the world.

2.5 The map and the territory: a direction of construction

The separations above fix a direction, and running it backward is the standing error of which reading every appearance literally is the local symptom. *The valid order builds the ontology from the evidence, the kinematics from the ontology, and the coordinates from the kinematics* [6]. The reification error runs the chain backward: it takes the coordinate system as given and reads an ontology off it, treating the map as the territory and an apparent symmetry of the description as a fact about the world.

Applied to a mature mathematical theory the demand is sharper than it first appears, because there the appearances include the formal structures one has derived. *A four-dimensional manifold that organises the record of events is a coordinate scaffold of this kind*; granting it existence runs the chain backward, and the discipline forbids it on the same ground it forbids reading the ecliptic as a real solar orbit.

And the reification can hide in something smaller than a manifold — in a verb tense. The founding literature calls a black hole a "collapsed" object, the past participle quietly reading horizon formation into the causal past of an exterior observer, where the causal structure places it only at the infinite-future boundary. *The grammar must follow the geometry* [2].

2.6 Requiring a phenomenon and permitting it

The last pair is a criterion rather than a distinction between things, and it is the one the rest of this paper is weighed on. *Prefer the world that requires the observed phenomena as a consequence of its structure to the world that merely permits them through adjustable parameters* [6]. A framework on which the phenomenon could not have been otherwise is an explanation; one on which it is a tuned possibility among many is a description.

In the ontological register this has a sharp form. A structure carrying an unforced parameter — a modulus fixing how some symmetry is broken, which neither the appearances nor any deeper principle pins — *is not a single world but a family*, and answers "what is the world?" with a family rather than a world. *It is inadmissible as it stands rather than merely disfavoured.* The maximally symmetric structure is the unique one carrying no such modulus: every less symmetric structure requires a choice of how to break the symmetry, and maximal symmetry leaves nothing to choose [6].

That clause is a statement about a group action and it has an exact dynamical counterpart. What leaves nothing to choose is a group acting *transitively*: a modulus is a coordinate transverse to the orbits, so one exists precisely when the action is not transitive. *Read on the substrate's geodesics the same transitivity says the flow is maximally superintegrable* — the isometry group acts transitively on the unit tangent bundle, so every geodesic is every other seen from another vantage [19]. *Least-arbitrariness and superintegrability are one property read epistemically and dynamically*, and a modulus appearing is the same event as the action ceasing to be transitive.

This is why §1's selection of the substrate is a selection and not a preference: the alternatives are not worse worlds but families of worlds, and the criterion that excludes them is the same one by which any theory predicting a datum is preferred to one accommodating it.

3 What is derived from the object

With the substrate fixed, the construction's working object is a *cut* of it, and one instrument generates the cuts. This section states what that instrument returns, what it cannot reach, and where its boundary falls — and it separates, at each step, the theorems from the reading they ground.

3.1 Straight cuts are vacuum

In the construction's gauge the vacuum condition $T_{\mu\nu} = 0$ is not a system to be solved but a single first-order linear ordinary differential equation, $rf' + f - 1 + \Lambda r^2 = 0$, whose *entire* solution space is

$$f = 1 - \frac{2M}{r} - \frac{\Lambda r^2}{3}, \tag{1}$$

the whole Schwarzschild–de Sitter family, with M the single constant of integration [12]. *The vacuum sector is exactly the kernel of the matter functional, derived rather than matched*: straight cuts are vacuum. Taken with the bend identity below — which shows every density is realised by some cut, so the functional is onto — the two give a kernel of dimension one over a vanishing cokernel, an index of 1, generated by that constant.

Two things follow that are worth having early. There is no room in the vacuum family for a third term: the solution space is the whole of it, so a rate written on this kernel carries Λ and the cut's offset and nothing else. And the question this answers is not a new one — it is the question the programme's own 2012 dissertation left open, asking what, if not the world-matter, fixes the geometry; the kernel is the answer, and it is one the corpus states in the asking author's terms.

3.2 Matter is the bend of the cut

Writing any curve as a departure from the vacuum profile, $f = 1 - 2m(r)/r - \Lambda r^2/3$, gives $8\pi T^t{}_t = -2m'(r)/r^2$: the energy density $\rho = m'(r)/4\pi r^2$ *is the radial growth-rate of the enclosed mass*, the bend of the curve off the constant-M profile [12]. Checked on a non-vacuum member, Reissner–Nordström–de Sitter is the bend $2m(r) = 2M - q^2/r$, returning the electromagnetic stress-energy off its q^2/r^2 term.

And the reading is not an artefact of the symmetry, which is the objection it would otherwise invite. For a general leaf the density is the leaf's intrinsic-curvature departure from the substrate, $16\pi\rho = {}^3R + K^2 - K_{ij}K^{ij} - 2\Lambda$ — *which is the Hamiltonian constraint*, the spherical identity being its symmetric case [12].

What the identity states is what the bend ***is****, not why a leaf bends as it does.* That is the matter content's own generative law — a dynamics for the curve itself, as against the ordinary leaf evolution that carries it — and the construction does not supply one. *The source names it as the deepest question the construction opens onto* [12], and it is carried on the frontier below.

3.3 What the instrument reaches, and the wall

The reach is a theorem: *the range of the operator is the symmetry-reducible sector of general relativity* [8]. The size of what is reachable is set by how much symmetry a class spends — a finite parameter family where the class reduces to ordinary differential equations, a functional family where it remains a partial differential equation problem — and the reachable sector is the constrained, non-radiative skeleton of the theory.

The boundary of that reach has a positive identity, and stating it the obvious way gets it wrong. The wall is the loss of isometry, and what lies past it is *the onset of free gravitational radiation*: the graviton's two propagating polarisations, read on the matter side as dynamical inhomogeneous sources [8]. *And the sector past the wall is not merely named*: the companion dynamics paper works a confined case there — a polarised wave on the de Sitter background, its unpolarised cut a wave map with a conserved handedness, and the wall itself a regular radiative boundary rather than a metric singularity, having neither a measure-collapse nor a curvature divergence [10]. *That is what makes the handoff checkable rather than asserted: the far side has a worked member. So the wall is a seam and not a defect* — it is where generation by symmetry hands off to ordinary inhomogeneous evolution, and the handoff is a feature of the classification rather than a failure of the instrument.

One thing this section does not rest on is worth naming, because it is the part that looks like the result and is not. Within a reachable class the operator's four data — leaf, lapse, shift, vantage — supply exactly the functions the general invariant metric admits, so the cut spans the class. *The source says plainly that this surjectivity is not the content*: once the cut carries the class's function count, the cut ansatz is the general invariant metric and spanning is near-tautological. *The content is the identification of the vacuum members as the substrate's own family — the kernel — and of matter as the bend.*

3.4 General relativity's constraint algebra is the substrate's grading

At the symmetric cut the substrate's algebra splits as $\mathfrak{so}(5,1) = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(4,1)$, and all three symmetric-space inclusions hold, $\mathfrak{m}$ not being a subalgebra. Under the correspondence between $\mathfrak{m}$ and the normal deformations and between $\mathfrak{h}$ and the tangential ones, *these are the hypersurface-deformation brackets term for term* [11].

So the algebraic shape that makes the Dirac algebra puzzling — two normal deformations bracketing into a tangential one — is exactly the shape of a symmetric-space coset: two coset directions bracketing into the isotropy. They are the same grading. *The claim is a recognition rather than an addition*: no structure is added to general relativity, and what is exhibited is that a pattern the theory already carries is the substrate's.

The coefficient that makes the algebra an algebroid rather than an algebra — the structure *function* standardly identified as the canonical root of the problem of time — is, on the symmetry-reducible reduction, the coset metric of that symmetric space [11]. *And the paper states at once that this is not a naive tensor equality*: the structure function is the Riemannian inverse spatial three-metric, the coset metric the Lorentzian five-dimensional form of signature $(1, 4)$, and what is identified is the *reduced* structure function on the symmetric-cut pattern with that coset form.

3.5 What is a theorem here and what is the reading

The construction's covariance claim is that general relativity holds *a single geometry* invariant under change of chart, while this construction holds *the substrate* invariant under change of *geometry*, with the slicing curve as the gauge object: *"many line elements, one geometry" becomes "many geometries, one substrate."*

That is a reading, and the source labels it as one. The theorems — the vacuum kernel, the bend-density identity, the lapse split, the cosmological geodesic, the second-ruling embedding — are computed and verified; *the covariance reading is adopted on their strength and explicitly does not follow from them as a corollary follows from a theorem* [12]. It is stated here at that weight and at no other.

4 What is computed, and what it is confronted with

The construction meets data in three places: a rate, a set of light-element abundances, and the microwave background's angular structure. Each is stated here with the register its source gives it, and the third includes a measured disagreement for which no mechanism is in hand.

4.1 The rate: one parameter, and one epoch

The count is smaller than it first appears, and where the smaller count comes from is worth following, because it is easy to inherit the larger one. The vacuum family of §3 does carry two — the substrate scale and the offset of the cut, and no more [12]. *But the cosmology does not sit on a general member of that family.* It sits on the one where the two horizon roots merge, and there the mass is no longer free: it is fixed by the substrate scale, so *both* the amplitude and the rate of the expansion law are set by that one scale and the whole scale factor is a single function of it [14].

So the rate has one parameter and one epoch. What remains after the geometry is fixed is where the present sits on that one curve — and the quantity conventionally written as the offset carries exactly that: it is the present areal radius *in units of* the geometry's own fixed length, the two horizon roots' merged value. *The denominator is fixed by the substrate scale and the numerator is the moment*, so the number is a clock reading and not a second geometric datum. That is the same thing the construction says of the density ratio directly: it is a reading of cosmic epoch rather than a driver of the expansion [14].

And the distinction is worth the care, because the weaker phrasing understates the claim. Two parameters neither of which is a content of the universe is a good position; *one parameter, plus the time at which one happens to look, is a stronger one* — and the second of those is not a parameter of the theory at all, since any theory whatever must be told when it is being read. The epoch is measured directly and calibration-free — without a distance ladder, without the microwave background, and without any density.

The familiar cosmological form is then a coincidence of form, and the distinction matters more than it first appears. Written in the fitted pair, the same rate reads as the Friedmann one, with the density parameters following identically. *That is a translation between parameter sets and not a decomposition into components*: the two readings agree on the function and share nothing beneath it. A reader who takes the agreement as a decomposition has imported the very thing the construction declines to assume.

4.2 Three levels, and why a mis-assignment is fatal

Every rate-bearing quantity in the construction sits on one of three levels, and the assignment is not a convention [14]. *The foliation stacking rate* is read leftward — the cut is primary and a density is the name of its bend rather than its cause — and it is empirically forced rather than modelled. *The leaf-level local dynamics* is the ordinary Friedmann readout with radiation gravitating normally: the same single geodesic, read *inward* as dust collapse where the first reads it *outward* as cosmology. *The projection* is what an observer reads.

There is no boundary in time between the first two levels. The distinction is kinematic and holds at every epoch, the decomposition exact, the two rates differing by the radiation term alone. *A quantity assigned to the wrong level is not approximately right*; it is a different physical claim, and §4.5 turns on a case where the two levels give opposite answers.

4.3 The abundances, and the miss that discriminates

The cooling leg *is* a standard big-bang nucleosynthesis — cooling through the window at the standard rate from a fully dissociated start — and a genuine multi-nuclide network integrated explicitly on that history returns $Y_p = 0.247$ against an observed 0.245, $D/H = 2.51 \times 10^{-5}$ against 2.53×10^{-5}, ^{3}He at 1.05×10^{-5}, and ^{7}Li at the standard several-fold over-prediction — *all jointly from the single inherited* η, at the baryon density the microwave background reads from the peak heights, with deuterium at -0.5σ and helium-4 at $+0.5\sigma$ [18].

And the lithium miss is the discriminating result rather than an embarrassment to be noted and passed over. It is the standard over-prediction, reproduced — which is to say the construction inherits the standard problem exactly, neither curing it nor worsening it. *A mechanism that changed the early history would have moved lithium*; this one does not, because the window and the rate through it are the standard ones.

4.4 The low-multipole floor, and why it does not discriminate

Because the distance slicing is exactly flat while the cosmological layers are a closed three-sphere, the photons are projected through the *flat* geometry while only the *source* modes carry the closed quantisation — *not the hyperspherical transfer of a literal closed universe*, which would deliver the lowest mode to the quadrupole and no deficit at all. The result is a *parameter-free* deficit below $\ell \approx 8$, bottoming at $\ell = 4$ and *not* at the quadrupole [14].

The register here is split and the split is the honest part: the existence, location and minimum are established; *the depth is open*, two transfers differing by up to a factor of two. *One thing that might have bled into that depth does not.* The projection across the lift is adiabatic for all but the lowest few harmonics and degrades to order unity exactly where the tower is coarsest — and the residual there is bounded rather than estimated, because the suppression falls monotonically with harmonic index and *the tower has no mode below the lowest, so its least-suppressed mode has no source beneath it.* Mixing can only carry amplitude into more-suppressed modes, and the ceiling on transmitted power sits at 7.9×10^{-8} [13]. And the feature does not discriminate between frameworks — it is a prediction the construction makes for free, not a test it passes and a rival fails.

4.5 The acoustic comparison, and a rejection on the temperature spectrum

The acoustic *scale* is met, and the corpus records what meeting it cost: it is an accommodation, and it is spent. *The peak structure is a separate confrontation, and it goes against the construction.*

The transfer is validated on its control, which reproduces a standard code to 0.14% *in the configuration this construction's own number is taken in, and both arms are converged in the wavenumber integral.* On that footing the construction returns a first peak at $\ell_1 = 206$ with $\ell_1/\ell_A = 0.6830$ against the sky's 0.7312 — *a* 6.6% *deficit in position* — and height ratios of 1.759 and 1.612 against 2.217 and 2.277, with two wavenumber cutoffs agreeing to every digit [14]. *And the deficit is neither of the two instrument faults a reader would reasonably suspect*: the wavenumber truncation and the projection path are both removed there, and the polarisation source pulls both arms down by comparable amounts, landing the control on the sky because it was above it and carrying this arm further below because it was already there — *nothing about the operation differing between the arms.*

The seam datum's own freedoms are measured rather than left as an escape. Across the seventeen readings admitted by a four-peak criterion fixed before the numbers, the first peak spans 148 to 228 — a factor of 1.541, and 1.118 across the numeric-phase readings alone — so *the position is a statement of the construction and not a free choice*, and the sky's value lies inside that span. The same holds separately for the spacing, the phase intercept and the alternation ratio. *But a span containing the sky on each statistic one at a time is much weaker than a reading that reproduces the sky, and there is no such reading.*

What the readings do carry is a correlation, and it is the sharpest thing the scan returns. Position and alternation move against each other across the datum, at Spearman $\rho = +0.782$ and $p = 2.1 \times 10^{-4}$: as the datum carries the peak up toward the observed value, the second gap turns from contracting to expanding. Of the readings landing the first peak within one grid step of the sky's, none contracts; of those that contract, every one sits at $\ell_1 \leq 212$. *So the seam datum can buy the position or the alternation and not both, and no mechanism for that trade is in hand.*

This is where the level rule earns its statement, because the deficit's most natural explanation turns on it. It is tempting to say that the acoustic modes re-enter above the plasma's onset, so that none of them is driven and the comb should be uniform for that reason. *That census is taken on the stacking rate, and the rule assigns the perturbations to the leaf*, which carries a radiation term: on the leaf the band containing the first peak enters the horizon *while radiation dominates.* Those modes are driven, and the driving is measured on *both* arms: removing it moves this arm's first peak by 134 multipoles against the control's 56, so the arm is driven 2.4 times as hard as the control and *overshoots* rather than falling short [16].

So the confrontation does not end in a residual. The full-spectrum likelihood scores both arms on the same 133 bins of the temperature spectrum with the amplitude fitted, and the control returns 2.10 in χ^2 per bin against this construction's 118.4: *a rejection of this arm's acoustic spectrum on that statistic, and by a wide margin. The configuration does not account for it* — it costs the control a factor of two against the noise floor, and this arm sits fifty-six times the control — and what produces the gap is the position deficit and the height ratios above, carried across every bin [14].

What is open is narrower than a mechanism, and sharper for it. The construction's own level assignment rules out the explanation that would have covered it, and the two instrument faults a reader would reasonably suspect are both removed: not the wavenumber truncation, and not the projection path, with the positions grid-converged. *So the question left standing is* *whether the deficit producing the rejection is physical or instrumental*, and those candidates are narrowed without being closed.

5 What is dissolved

A dissolution is not a solution. Where a solution supplies a mechanism that answers a question, a dissolution shows the question rested on a premise the construction does not carry — so it is discharged rather than answered, and nothing is owed in its place. The cluster below is presented *graded*, because the tiers differ in what an objector must dispute, and a flat list would hand a reader the whole set to reject at once [16].

5.1 Tier one: what carries the layered reading's full weight

These stand or fall with the reading of §1, and an objector who declines that reading declines these with it.

The horizon–singularity family. The event horizon of a collapsing body is a metric singularity met only at infinite exterior time, so no completed horizon is realised on any finite layer. Three results follow, and each is discharged for a different reason.

Penrose's theorem is a correct mathematical result whose hypothesis is a *realised* closed trapped surface interior to the event horizon. With no completed horizon there is no trapped surface it would enclose. *Singularities are not avoided by new physics*; they are rendered physically irrelevant by causal structure alone, the curvature singularity remaining a feature of a global extension the realised worldtube never instantiates — and cosmic censorship is correspondingly unneeded rather than proved [2].

The Bogoliubov construction behind horizon radiation requires a globally defined horizon, a completed causal structure joining past to future null infinity across it, and permanent loss of causal contact rendering the two vacua inequivalent. *The third has an exact criterion*: two Fock representations are unitarily equivalent precisely when β is Hilbert–Schmidt, and a thermal β fails that test at the *infrared* end, its $1/\omega$ tail making the norm logarithmically divergent [2]. *This is not the claim that black holes do not radiate*; it is the claim that the construction's third premise is unmet on a realised worldtube.

The information paradox arises only if a completed horizon forms and subsequently evaporates. *Both premises fail together*: no completed horizon forms, and with no horizon-induced radiation there is no evaporation to carry the loss. The realised spacetime remains globally connected, with a global Cauchy surface, no hidden interior sector to trace over, and unitary evolution unobstructed. *The paradox is not resolved by a mechanism recovering information — it does not arise* [2].

And the criterion that sorts these is completion, not perspective. An argument of the form "the horizon is perspectival, therefore no thermal flux" is refuted before it starts and is not the argument made: the Rindler horizon is observer-dependent *and complete* — the boost field an exact Killing field of the whole spacetime — and the Unruh spectrum is thermal. Across the four horizons the programme names, one criterion sorts all four, and it is completion [2].

The problem of time, the hole argument, and closed timelike curves sit in this tier for the same reason: each turns on whether a foliation is a convention, and the augmentation of §1 makes it a physical structure. Where the foliation is physical, the canonical frozen-time problem is a question about a convention rather than about the world; the hole argument's two solutions are the same world read twice; and a closed timelike curve is not a trajectory an existing layer can carry.

5.2 Tier two: what follows from the single scale alone

These require less. *An objector may reject the layered reading entirely and still owe an answer to them*, because they follow from the substrate carrying one scale and nothing further.

The fine-tuning pair — the cosmological constant problem and the coincidence problem — is discharged by the accounting of §1: Λ is the substrate's waist rather than a vacuum energy to be computed and matched, and the constants through which physics is written on it are unit gauges. *The question "why is the vacuum energy so small" presupposes that the quantity is an energy*, which is the premise the one-scale reading does not carry.

The local–cosmic boundary is the second: the locus where the local bend exactly cancels the substrate's cosmological term is derived rather than posited, so *the threshold separating what expands from what stays bound is a length the construction computes*, not a density it fits. *It is the subject of §6, which sets out why the same radius carries several unrelated readings.*

5.3 The cosmological puzzles

These are a separate family with a separate home, and they are not graded by the tiers above — they follow from the rate of §4 rather than from the layered reading.

The reckoning is drawn plainly at source: the standard account assembles dark energy, inflation for causal contact and flatness and coherence and scale invariance, and further physics for the tensions; *the construction reads each off* Λ [14]. The horizon and flatness problems are not solved by an early mechanism but do not arise on a rate whose uniformity is forced rather than assumed. The big-bang singularity is a coordinate feature of the reading rather than a breakdown of the theory, the minimum radius being finite. The coincidence problem is dissolved by the same accounting that discharges the fine-tuning pair.

And the oldest of these is the one the construction answers most directly, so it is stated first rather than folded into the list. In any model whose expansion is driven by its contents, matter and radiation decelerate it, so such a universe expands at late times only because it was already expanding, and the expansion is an inertial legacy of an initial condition at which the description breaks down. *Eddington put the objection plainly and Hoyle shared it*: postulating that the velocities were there from the beginning can scarcely be called an explanation of them.

On this construction the acceleration is not a sector at all: it is the existent slice's own intrinsic curvature, so the expansion decelerates while that curvature is negative, turns over exactly where it vanishes, and accelerates thereafter — *with nothing added to make it do so* [14]. *The identity is set out in §6, together with the reason the locus of that turnover is also, for every mass, where a bound structure's hold gives way to the flow.*

The coincidence problem goes with the same accounting. The present matter density and the dark-energy fraction are *readings of one cosmic clock rather than drivers of the expansion*, so the coincidence of the two dissolves into our observing at a time of order the geometry's one timescale [14].

And the Hubble tension is placed in a different category from the rest, which is the honest part. It is carried not as a dissolution but as a *discriminating datum* — the standing here being theory-choice-favoured rather than settled [14]. *A reader should not count it with the others.*

5.4 The unification divides, and the matter sector

Two further families are carried at lower altitude than anything above, and the corpus says so.

The gravity–gauge and general-relativity–quantum divides are addressed where the boundary paper draws its synthesis: the pieces are established, and what is not discharged is *the identification* — that the substrate's two real forms are physics' actual quantum and gauge sectors rather than a structural rhyme with them [17]. *That is a located question and not a want of confirmation*, because its discriminator is named: a rhyme predicts nothing further, an actual gauge sector carries a field strength and a coupling, and the flat bundle of §7 is exactly what cannot supply one.

The matter-sector family — generation replication, the origin of colour, the family-by-chirality structure, the mass hierarchy, matter and antimatter, CPT — is addressed in the matter sector's own scope section: generation replication read as the three hinges, the family symmetry as the substrate's own Weyl group, gauged chirality against global flavour as the on-versus-tangent distinction, and matter against antimatter as the conjugation [7]. *These are held at forced-within-CR coherence with the content external, and the corpus records this as the lowest altitude it carries.* §7 states what the same sector declines, and the two should be read together.

5.5 The closure: one substrate read many ways

The meta-collector, where the rest close. The substrate's maximal symmetry is worn seven ways, and the physics is read as broken-symmetry shadows of one object [19]. *Its status is stated in §1's terms*: the seven results are each established, and *their unification is the thesis rather than a further theorem* — a conjecture decidable by a named test, and carried here at that weight.

5.6 Tier three: what stands on the bare analytic structure

This tier is untouched by any verdict about what exists, which is what makes it the most robust and the least often noticed. It requires neither the layered reading nor the one-scale accounting — only the analytic structure of the solutions.

The continuation through the curvature singularity is of this kind, as is the identical analytic type of the two critical points. *And the Kretschmann divergence is a pole of finite order* — twelfth in the cycloid parameter, raised from sixth by the chain rule — *which is the whole of why the continuation is available*: a pole is continuable where an essential singularity would not be [3]. The overcritical regime is reached by the same continuation that joins the seam.

An objector who grants nothing else still faces this tier, and that is the reason the cluster is graded rather than listed. Rejecting the reading of §1 costs tier one and leaves tiers two and three standing; rejecting the one-scale accounting costs tier two and leaves tier three. *There is no single objection that takes all three.*

6 One radius, read seven ways

This is the corpus's clearest convergence, and it is worth setting out whole because no single paper contains it. A boundary observed around every bound structure, and the epoch at which the universe stops decelerating, are on this construction *the same fact at different masses* — and the geometry that makes them one is visible on the substrate itself.

6.1 The locus

For the marginally-bound congruence that *is* this cosmology, $(\mathrm{d}r/\mathrm{d}\tilde{\tau})^2 = E^2 - f$, so

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\tilde{\tau}^2} = -\frac{f'}{2} = r\,K_G, \tag{2}$$

with $K_G = 1/\alpha^2 - M/r^3$ the slicing surface's Gaussian curvature: *the comoving acceleration is the slice's own intrinsic curvature*, up to the positive factor r [4, 16]. Both sides vanish at

$$r^3 = M\alpha^2, \tag{3}$$

generally in the mass. That radius is the Hubble–Eddington radius, $r_{\mathrm{HE}} = (3GM/\Lambda c^2)^{1/3}$.

So one equation says two things that are not obviously about each other. The expansion decelerates while the slice is negatively curved, turns over exactly where the slice is flat, and accelerates thereafter. And that same flat locus is *the boundary within which local structure stays bound against the cosmic flow and beyond which it is carried off* — the radius the largest bound structures are observed not to exceed [14].

6.2 Why this is a convergence and not a restatement

Two independent quantities change sign at that radius, and the construction does not relate them. The slice's intrinsic curvature is geometric; the acceleration of the areal radius is dynamical; *nothing connects them beyond their both being read off the same curve* [6]. A geometric identity restated in dynamical words would be one fact twice. This is not that.

And the corpus reads the radius seven distinct ways, in papers written for unrelated ends: the Hubble–Eddington radius proper; the flat locus of the slice's own curvature; the handover from the sub-marginal bound orbits to the marginally-bound congruence that is this cosmology; the local range of the substrate's single scale, the one Λ that also sets the global expansion; the acceleration turn; the unique fixed point of the vantage involution [9]; and a consequence of the layered closure itself [6, 16]. *No single paper lists them all* — five are gathered in one place and five in another, three of them shared — *and the convergence is visible only from outside any one of them.*

6.3 The picture on the substrate

Drawn on the hyperboloid the two behaviours are one curve. The substrate is finite at its equatorial throat and opens toward the poles. A slicing line at offset r_0 meets the throat circle rather than passing through it — the manifold is not there at the centre — so the only continuation available is around the circle, and the chart's $r = 0$ sits on the throat at the back, diametrically opposite the hinge [4]. *The lap runs around the throat and then out along the horn.*

The rate along it separates the loci exactly. Since $(\mathrm{d}r/\mathrm{d}\tilde{\tau})^2 = 2M/r + r^2/\alpha^2$, the rate vanishes at the comoving turnaround, diverges as $r \to 0$, and *passes a minimum at* $r = (M\alpha^2)^{1/3}$ — the locus of (3) [16]. *So the re-expansion is slowest exactly there*, and what (2) adds is that the same point is where deceleration becomes acceleration.

On the forced member the locus coincides with the front seam, where the horizon cubic's double root makes f and f' vanish together; there the bead attains $\mathrm{d}r/\mathrm{d}s = 1$ with $\mathrm{d}^2r/\mathrm{d}s^2 = 0$, its unique inflection. *On other masses the two separate with the roots* [16, 4] — *which is why the radius is a statement about every structure and the seam is a statement about one.*

6.4 What this dissolves, and what the standard account has instead

The observed phenomenon has a name and a long-standing awkwardness. A galaxy at the edge of its cluster is said to "enter the Hubble flow" and be carried off, and the cluster to evaporate at its boundary. *In an account where the expansion is driven by the contents, there is no mechanism for that edge* — only a balance between an attractive term and a repulsive one, meeting at a radius where their magnitudes happen to agree. *A coincidence of forces is a place, not a reason*, and the picture of a structure shedding its outskirts across a threshold has had no better ground than that for a century.

The radius itself is not new and the corpus does not claim it. It is derived in the standard framework and proposed there as an independent local test of Λ [5], with the same $M^{1/3}$ scaling. *That literature calls it a maximum "turnaround" radius, and the word is used here only in reporting their result. In this construction* turnaround *is a different locus and the two must not be run together*: the comoving turnaround is where the *worldline* turns, the radial kinetic term vanishing and the collapse dynamically reversing, at $r = -(2M\alpha^2)^{1/3}$ on the conjugate leg [16]. *The radius of this section is where the* acceleration *changes sign, at* $r^3 = +M\alpha^2$, and it is called the Hubble–Eddington radius throughout. *What differs is what the radius is.* On this construction it is not where two terms balance but *where one quantity changes sign* — the slice's own intrinsic curvature, whose sign *is* the boundedness–expansion dichotomy. *So the threshold is sharp because a sign is sharp*, and the structure does not shed its outskirts against a competing pull; the curve those outskirts are carried along simply turns.

6.5 What the appearance is a shadow of

Nothing accelerates anything at that radius. The sign of $\mathrm{d}^2r/\mathrm{d}\tilde{\tau}^2$ is a property of a fixed curve, read where f' passes through zero, and the deceleration and the acceleration are what continuous motion along that curve looks like from within [6].

So the boundary's appearance — a coincidence of forces, a structure's hold giving way to a flow — is shown what it is a shadow of, and not merely declared to be one: a curve whose second derivative changes sign, seen by an observer carried along it. *That is the reading of §2 carried out on an observed phenomenon*, and the phenomenon it explains is the local–cosmic boundary itself.

One further reading marks what the coincidence is not. On the forced member that radius is the front seam, which the stratification marks as the one locus crossed at unit speed *without* dividing two causal regions — f touching zero without changing sign. *So the areal acceleration changes sign there and the causal character does not*: two independent sign-questions answered oppositely at one radius, and an observer crossing it records a change in their expansion history and none in the character of their own radial coordinate [6].

7 What it declines to claim

The declines below are not gaps awaiting work. Each is a place where the construction has an argument for why the thing it does not deliver *cannot* be delivered on the structure it has — and stating them with their registers is what keeps the deliveries of §§3–5 legible as deliveries.

7.1 Colour is not a geometric isometry, and the obstruction is complete

The substrate's continuous symmetry cannot carry the strong force, and the reason is exact: $\mathfrak{su}(3)$ is not a subalgebra of $\mathfrak{so}(5,1)$, *precisely because the isometry is exhausted* [17]. *Every candidate substrate bundle falls with it*: the algebra needs a complex rank-three module, and a real bundle's complexification carries a parallel conjugation whose holonomy lands back in the real form.

So the question was never which real bundle, but where the complex structure is — and it is at the branch point. Taking the wall's definition literally there are three branch loci, one per vantage, and the three wall monodromies with the hinge three-cycle generate $SU(3)$ itself, a full lap being the centre [7].

What that delivers is colour's discrete content, and it is delivered. What it does not deliver is a force. *The flatness is a complete obstruction rather than a stage not yet reached*: a flat bundle has no curvature to be a field strength, and the corpus shows this rather than conceding it [7]. *A reader should not expect a later section to supply the gauge dynamics; there is an argument here that it cannot be supplied on this structure.*

And the antisymmetry that makes the baryon work comes from the same discipline: it is supplied *not by a gauge datum* but by the three modes being the kernel of one Dirac operator — hence identical particles, so a three-fermion state lives in Λ^3 and *the surplus invariants do not exist. The diquark is then a prediction rather than a fit*, nothing having asked that the two-fold carry no invariant.

7.2 The count is computed; the hierarchy is not

The generation number is three, as a γ^5-graded index of wall-localised zero modes [7]. *The count is computed rather than traced*, and it is the sector's strongest single result.

The mass hierarchy is not delivered, and the corpus states the boundary rather than blurring it. A cubic supplies the *count* and not the spacing: three roots are three, and nothing in the count fixes the ratios between them. *So a reader who takes the generation result as a step toward the mass spectrum has crossed a line the source draws explicitly.*

7.3 What is held open, and at what altitude

The identification of the substrate's two real forms with physics' own quantum and gauge sectors is not discharged, and this is the decline that matters most for how the rest is read [17]. *It is declined for a stated reason rather than pending evidence*: what separates an identification from a structural rhyme is dynamics, and the flatness obstruction above is precisely what denies the compact face a field strength. *So the decline names its own blocker*, which is what distinguishes it from a promise to confirm later. The matter-sector family of §5 is carried at *forced-within-CR* coherence with the content external — the corpus's own description, and the lowest altitude it assigns to anything.

And the closure of §5 is a conjecture and is labelled one. The seven ways the substrate wears its symmetry are each established; *their unification is the thesis*, decidable by a named test and not established by the seven results individually [19]. *Asserting the unification on the strength of its parts would be the one over-claim this corpus is most exposed to*, and the grading of §5 exists partly to prevent it.

7.4 Why the declines are discipline rather than modesty

Each decline above buys something. The colour obstruction is what makes the discrete residue the single geometric opening — had the continuous symmetry been available, the residue would be one route among several rather than the only one. The hierarchy boundary is what lets the generation count be read as an index rather than as a fit. And holding the identification undischarged is what lets the structural results be stated at all without asserting more than the construction shows.

These are not places the corpus stopped; they are places it drew a line and then used the line.

8 The joins

The convergence of §6 is one kind of join: several papers describing what turns out to be one fact. There is a second kind, and it is what an assembly can supply that a set of papers cannot. Twenty-three mathematical fields were put against all seventeen papers — what bit, what bounced, and where the boundary lies — and the identities below are ones *the papers state separately.*

8.1 Identities the papers state separately

The Nariai condition and Petrov type D are the same algebraic event. The eigenvalues in the classification result are the self-dual Weyl operator's, and its degeneracy condition is Nariai's own algebra: *a depressed cubic's discriminant vanishing*, on two different cubics.

The ellipse of the slicing construction has the Killing form's eigenvalues, its axis ratio the A_2 root-to-weight ratio, and its shorter semi-axis the slicing scale itself.

The S_3 is worn three ways: as the monodromy group of a three-sheeted cover, as the Weyl group of A_2, and as the Galois group of the horizon equation over $\mathbb{C}(2M)$ [9].

Colourlessness has two halves and they come from different places: the trivial Fourier summand on the deck $\mathbb{Z}_3$ supplies the *necessary* half, the antisymmetry in the triple tensor product the *sufficient* one.

Matter is the obstruction to integrating a connection, not a quantity standing beside it — which is the bend of §3 read in the language of the field that owns obstructions.

And the peak of the crossing is an erasure channel, so what it returns is exactly what a conservation law protects: the inherited datum survives where the composition does not, in one sentence.

8.2 Two halves of one index, neither of them in the papers

The generation count of §7 needed two things established that no paper had established. That the wall index is well *defined* — the leaf compactness surviving the limit — came from one field bake, *which the corpus's own receipt had never tested.* That it is *stable*, and that a spectral gap is why, came from another. *Two bakes, two halves, and neither half was in the papers.*

An open join is named with them: the construction relies on two spectral gaps doing two different jobs, with neither referencing the other. *That is carried as an open join rather than assumed to be one gap.*

8.3 What the instrument's negatives are worth

A field that bounces cleanly is a measurement, and most of the recent bakes bounced. The apparent probabilistic footprint of the corpus resolves, on inspection, into a curvature, an invariance and a substring; *congruence* is geodesic throughout and never arithmetic. *Reporting that is what makes the joins above worth anything* — an instrument that only ever finds connections is not measuring.

And the instrument checks itself, which is the strongest evidence it works. A spectral enquiry asked whether a mode degeneracy used elsewhere matched the corpus's own and found that they differed: *the papers derive one expression where the textbook supplies another, and the papers' is the one this construction requires. The disagreement was found by one field bake reading another*, which is a check no single bake can run on itself.

8.4 A debt that is presentational and not scientific

One class of finding is worth stating because it cuts in exactly one direction. A method can be used correctly throughout a corpus and named nowhere — and *a reader searching for the standard name finds nothing and concludes the method is absent, when it is present and correct.* Five such were carried; four proved already named on re-measurement, and the fifth — *matched-procedure differencing*, where a control is run through the identical extraction so the procedure's own bias cancels in the difference — is named now.

The class closes silently, which is why it goes stale: someone writes the name while doing something else and nothing records it. *A naming debt wants re-measuring rather than re-reading*, and that distinction is the reason this one shrank from five to one when it was finally measured rather than recalled.

9 The frontier, at its true relative size

What follows is complete and unfiltered. Eight items stand open, each with what would discharge it, and none is omitted for being awkward or small. *The list is given whole because a filtered frontier is worth nothing*: a reader cannot weigh a programme against a selection of its own open problems, and an item that leaves a list by reclassification rather than by being worked has not been closed.

The relative size is a number rather than an adjective. Eight open items stand against roughly seventy-four results carrying a hard register — theorem, proposition, derived, computed, receipt-anchored. *The ratio is not the argument*, and it is given because the alternative is an adjective, and an adjective about one's own frontier is the thing a reader is right to distrust.

Assembling this paper opened three items and closed all three, which is worth stating because it cuts both ways. Each was raised where the construction seemed to state an opening and no instrument was tracking it; each turned out, on being read at source, to be answered in the paper it came from or by material the corpus already held unjoined. *A frontier that moves when a corpus is read whole is the reading working* — in both directions, and a list that only ever shortens is no more trustworthy than one that only ever grows.

9.1 The matter sector

The deepest item the construction opens onto is the matter content's own generative law: a dynamics for the curve itself, as against the ordinary leaf evolution that carries it. §3's identity says what the bend *is* and not why a leaf bends as it does. *Discharged by* a generative law exhibited.

Whether a fermion sector can be built on the compact face stands with it, and it is well posed in a way few unbuilt things are: bounded by a structural result fixing where an admissible mechanism may live and by measured ratios fixing what it must produce, *with its motivation settled and no ontological claim riding on the outcome* [17].

And one question this sector might be expected to settle is one it demonstrably cannot. The per-generation content is fifteen Weyl fermions, or sixteen with a right-handed neutrino. *Both of the construction's handles on matter content are blind to that state*: the generation count is a graded index of wall-localised zero modes and counts generations rather than the content within one, and the anomaly condition returns the identical verdict for fifteen and sixteen, the singlet carrying no colour, isospin or hypercharge and contributing exactly zero on every channel [7]. *So a preference between them must come from outside the construction*, which is the representation-content step §7 records it as declining.

9.2 The quantum sector

The ultraviolet definition of the mode sums is open and has never been attempted; the corpus records two of its three parts as settled and names the third. *And its standing is worth stating with it, because it bounds what an unresolved item costs*: it is the same wall any interacting quantum field theory meets, and the classical dynamics, the deparametrized unitary evolution and the parameter-free closure of the quantization ambiguity *stand independent of its resolution* [13].

And the propagating nonlinear residual is worth reading closely, because it shows what an exact result not reaching something does and does not mean. The ghost and zero-mode runaways are closed *exactly*, and this is the one structure those results do not reach — *but it is not therefore open.* It is the future stability of de Sitter, and the small-data theorem covers precisely the perturbative regime the propagating graviton lives in, with the all-data extension in the same symmetry class corroborating and the one non-generic exception falling exactly on the locus the range classification already isolates. *And the reason no energy instrument was ever going to settle it is given*: the method trades the global-in-time problem for a local one at future null infinity, so *the failure of any monotone energy is the expected shape of that fact rather than a sign of openness.*

9.3 The cosmology

The acoustic rejection of §4.5 is the largest open item in the paper, and it is open in the strongest sense. *The transfer that measures the deficit and the likelihood that scores it are both run, and the likelihood goes against the construction* — the numbers are given at the confrontation rather than repeated here. *So what this list carries is not a residual awaiting a mechanism*: it is the question that section leaves standing, whether the deficit producing the rejection is physical or instrumental, with the two named instrument faults removed and the positions grid-converged.

One item here is a test rather than a gap. The Hubble–Eddington radius of §6 is degenerate between this construction and the standard one, and *the circularity is in how the test has been run*: the calibrations are dark-matter-only and the mass estimates dynamical, so which mass the radius tracks is assumed rather than measured. *Discharged by* the radius compared against independently measured baryonic mass.

And one thing stood here as a numerical curiosity and turns out to be a consequence. The normalisation of the standard linear growth factor crosses unity once, at the concordance matter density — an observation recorded from inside the standard model, with no explanation attached. *Written on this construction's rate the growth equation loses its parameters altogether*: the substrate scale cancels, because on the forced member the source is fixed by that scale while the time variable carries it too, and what is left is a single equation in the clock with nothing free in it. *The decaying mode is the expansion rate itself*, and the matter fraction is not a parameter but sech^2 of the clock [14].

On those variables the condition that the normalisation equal unity reduces to a balance: the growth-weighted history of the matter fraction, taken about the value $2/3$, integrates to zero. *The pivot is the deceleration-to-acceleration turnover*, $\rho_m/\rho_\Lambda = 2$, and the balance is solved at the observed density [14].

Two things narrow that, and both are worth having. The turnover is not this construction's property — any flat matter-and-Λ rate has one. What *is* this construction's is that the turnover sits at a geometric locus, the radius of §6: *the balance pivots on the turnover, and the geometry is what makes the turnover a place.* And the rate the derivation runs on is the stacking one, which is right for the object — the published normalisation is a matter-and-Λ integral and the stacking rate written in the fitted pair *is* that rate — but the construction assigns perturbations to the leaf, so *what is derived is the root of the standard normalisation rather than this construction's own growth root.*

So a number recorded from inside the standard model as a curiosity is, on these variables, fixed by an equation with nothing free in it, and what stays contingent is that the present epoch lies near it — a question about when we look rather than about what the world contains.

9.4 The substrate geometry

The identification of the flat synchronous space with the substrate's second ruling is the sharpest statement in the corpus's $r = 0$ chain, and it is bounded to the cosmological face by the orthogonality it requires — a collapse horizon's limiting direction being generically non-orthogonal to any spacelike slice [12]. *Whether a charged collapse forms the Cauchy horizon* is open, and it is a question about general relativity's own dynamical interior rather than about this construction — *the eternal geometry's obstruction is stated exactly, and a collapse is a dynamical thing. What is at stake in it is bounded*: the closed-loop reading of the charged case, and not the result that collapse continues as a cosmology, whose hypotheses name no charge and which a sub-extremal charged collapse satisfies [4]. *Whether $-M/r^3$ is a forced pivot or a mass* awaits an adjudication. *What survives for a perpetually collapsing body is not on this list, and the reason is worth stating*: the horizon-thermodynamic apparatus, area law and entropy alike, has on a finite layer no realised horizon to be defined on — which is a result and belongs with §5 — while what the collapse produces is the central theorem's, and what content crosses is the erasure channel's of §4.

9.5 The eight, with what would discharge each

Stated as a list so it can be worked rather than read.

open item	*what would discharge it*
the matter content's generative law	a dynamics for the curve itself, exhibited
a fermion sector on the compact face	the sector built, or an obstruction shown — its motivation is settled and no ontological claim rides on the outcome
the ultraviolet definition of the mode sums	the definition attempted
the acoustic deficit — a rejection on TT shape, not a residual	whether the deficit is physical or instrumental; the transfer and likelihood are run and the two named instrument faults are removed
the derivation of the inherited datum	a conserved charge of the progenitor exhibited
the onset, and the progenitor spectrum	a progenitor interior modelled
which mass the Hubble–Eddington radius tracks	the radius measured against independently measured baryonic mass, rather than against the dynamical mass the calibrations assume
the charged collapse and the Cauchy horizon	whether a dynamical charged collapse forms it — a question about general relativity's interior, with the cosmogenesis untouched either way
the attribution of $-M/r^3$	an adjudication between forced pivot and mass

Nine rows for eight items: the onset and the progenitor spectrum are carried separately because they have different fates — *the onset ratio's own derivation is closed by impossibility* (§5) while the spectrum is ordinary open work, and a single row would hide that.

9.6 What is not on this list, and why that matters

Two kinds of thing are deliberately absent. Items closed by impossibility are not carried as open: the onset ratio changes along the leg, so a quantity with no single value has no handover to transmit it, and what stands is not a derivation owed but the statement that there is none to give [19]. *An impossibility is a result and belongs in §5, not here.*

And the declines of §7 are not frontier items. The colour obstruction is complete rather than a stage not yet reached; the mass hierarchy is a boundary the construction draws rather than a gap it has failed to close. *Listing a decline as an open problem would misreport both*: it would promise work the construction has argued cannot be done on this structure, and it would inflate the frontier with items nobody owes.

10 What would displace it

Two questions are asked of a framework carrying a measured disagreement of this size, and they are not the same question. The narrow one asks whether the disagreement is an instrument defect. The wide one asks what would disqualify the framework, and it is not a question about that disagreement at all.

10.1 The narrow answer, kept short so it is not mistaken for the wide one

For the acoustic rejection of §4.5 to be a fact about the framework rather than about the instrument, there would have to be a physical defect that suppresses the third peak's ratio several-fold, leaves the peak positions right to within a few per cent, leaves the acoustic angle right to within two standard deviations and free of the Hubble constant, *and produces the identical failure on the standard model, where the answer is independently measured. The last* *requirement is very nearly a contradiction.* Every candidate examined so far that moved a number was an instrument fact, and none required a physical claim to give way.

10.2 The wide question

What retires a physical framework is a better or equivalent account of what it explains. Nothing else ever has. So the price of disqualifying this one is the register in full — and the register is not an illustration but a count: nine structural recoveries, the dissolutions of §5, the cross-field joins, a formal spine of some two hundred environments with fourteen named theorems on six axioms, an evidence rail of several hundred receipts of which the overwhelming majority are shown able to fail, and forty-four proved classical-geometry identities.

Every one of those is a thing a competitor must also explain. The Carter constant is one: general relativity carries it as a coincidence of the Kerr solution and this construction derives it, with a separate result supplying why the derivation is not circular [8]. *It is an illustration of the class and not the case* — the same paragraph could be written for the provable absence of the radiative types, the generation index, the group obstruction, or the index-one vacuum.

10.3 On parameters, where a deficit is easily manufactured

It is tempting to list the framework's measured quantities — the inherited datum, the amplitude, the tilt — as unpaid debts because they are measured rather than derived. That is not a debt: it is what a physical theory does with parameters, and the comparison class does the same. The standard model fits six; this construction fits one, with the background otherwise two numbers, one of them measured calibration-free (§1).

A framework that fits fewer parameters than the model it competes with does not owe a debt for fitting any, and presenting that discipline as backlog is an error worth naming rather than quietly avoiding.

10.4 What is genuinely unpaid, at its actual size

Two real builds: the full-spectrum likelihood and the end-to-end transfer, both of which the cosmology names itself. *One unresolved observational question*: the high-multipole damping signature, established as real and non-reabsorbable but not established in what it does to the observable, being largely degenerate with the spectral tilt at fixed acoustic angle. *And two items that belong to the field rather than to this framework*: the lithium over-prediction, which is standard nucleosynthesis's and is recorded because it sits in the sector rather than because it is owed; and the horizon-entropy convention, adopted rather than derived — *a convention shared with all of black-hole thermodynamics*, with the further claim correctly declined.

That is the honest unpaid column, and it is short because the register is long — not because anything has been left out. §9 lists the eight open items in full, and this section is the weighing rather than the list.

10.5 The claim this supports, stated exactly

A framework with this register is not immune to disqualification. It is expensive to disqualify. The two builds above are where a competitor would start, and they are named here at the same weight as the recoveries — because a register offered as a defence, with its own weakest points withheld, would not be worth the counting.

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Appendix L The Knowledge Ledgers

Each claim marked L in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

`QUADRIC_GEOMETRY_LEDGER.md` [field bake]

projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes